

FORECASTING FILM INDUSTRY GROWTH: ECONOMIC IMPACT AND AI-DRIVEN DECISION SUPPORT IN NEW JERSEY



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Abstract

This study evaluates the economic impact of New Jersey's expanding film and television industry using forecasting, comparative state analysis, and applied artificial intelligence (AI). The project examines how production growth, supported by New Jersey's 2018 film tax incentive policy and major studio investment, is associated with employment, wages, and industry establishments. We analyze a 20-state comparative quarterly time series dataset covering 2006 through 2024. The dataset combines Bureau of Labor Statistics Quarterly Census of Employment and Wages data for NAICS 5121, the classification for motion picture and video industries, with state GDP, unemployment, inflation, production counts, tax incentives, and film spending data. New Jersey is compared with 10 incentive states and 10 lower-activity non-incentive states. An interview we conducted with an assistant director highlights how New Jersey's tax incentives helped attract major studio investment, including Netflix's planned Fort Monmouth project, and strengthen long-term production infrastructure. Trend analysis shows steady wage growth in New Jersey's industry, with average weekly wages increasing by approximately \$54.64 per year ($R^2 \approx 0.79$) and total annual wages increasing by approximately \$15.76 million per year ($R^2 \approx 0.80$). Post-2018 models indicate stronger industry development, including higher weekly wages and a shift from declining establishments before 2018 to increasing establishments afterward. Forecasting projects New Jersey production spending to increase from approximately \$874.7 million in 2025 to \$1.12 billion by 2030, supported by a regression model that explains substantial variation in spending ($R^2 \approx 0.83$). The study also introduces Location Scout, an AI platform we created to identify high-potential filming and development areas in New Jersey. By translating forecasts into location-based recommendations, Location Scout turns economic modeling into a practical decision-support tool for sustainable industry growth and regional development.



Background Information

New Jersey's 2018 film tax incentive expansion made the state a highly competitive production hub. Beyond direct hiring, this activity generates massive economic spillovers for local vendors, hospitality, and municipal services.

Rapid growth often concentrates exclusively in already-established filming hubs. As a result, newer, officially trained Film Ready Communities need to gain visibility with major producers.

NJ officially trains and certifies towns to be film production ready. These communities are equipped with streamlined permitting, dedicated local liaisons, and the exact infrastructure needed to support major filming logistics.

NJ Film Ready Location Scout acts as a sustainability-focused decision-support tool. Using ethical AI design, it connects production needs directly to community assets, ensuring overlooked towns have a fair chance to attract statewide film spending.

Methods

Data: 20-state film industry dataset, 2006–2024, with 380 state-year records and 7,600 data points

Source: BLS QCEW NAICS 5121 film and television industry data

Measures: Employment, total wages, average weekly wages, establishments

Controls: GDP, unemployment, population, population density

Analysis: Interrupted time-series, state comparison, synthetic control, forecasting

AI Output: NJ Film Ready Location Scout for ethical and sustainable location recommendations

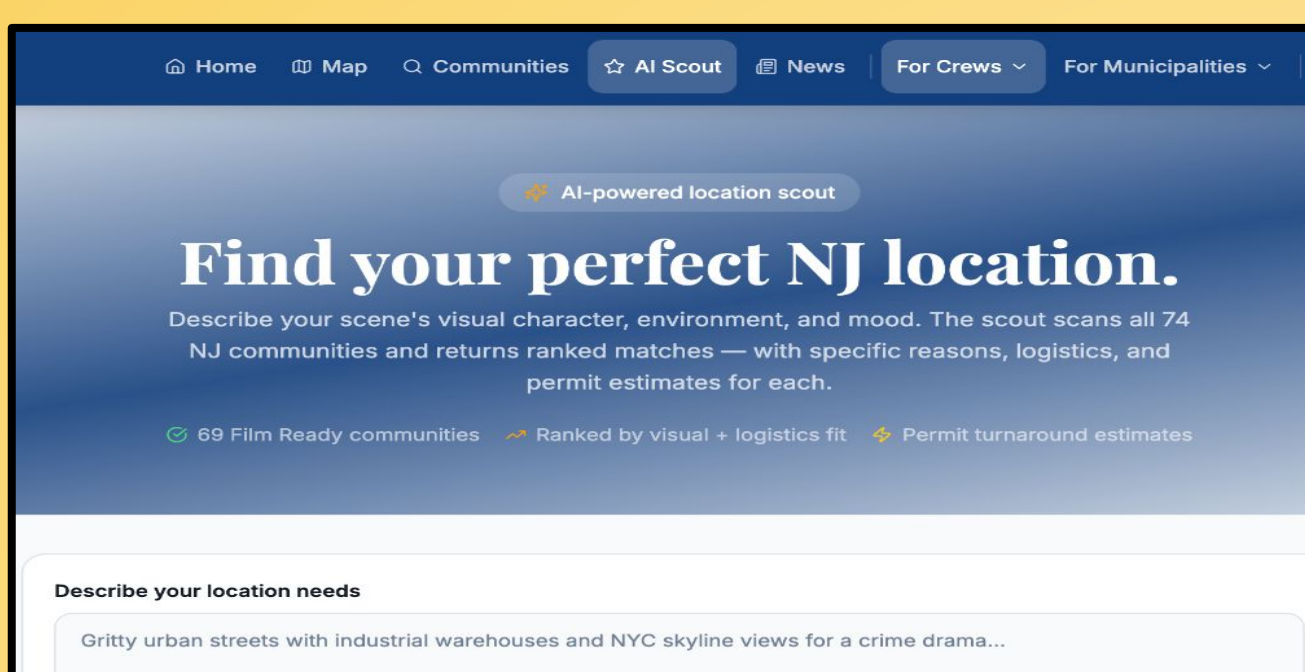
Results

- Average weekly wages increased by **\$277 after 2018** and **\$349 after 2020** ($R^2 \approx 0.78$).
- Establishments increased by **64 after 2018** and **162 after 2020** ($R^2 \approx 0.92$).
- Total wages increased by approximately **\$14.11M after 2018** ($R^2 \approx 0.71$).
- Synthetic control showed NJ's post-2018 weekly wage was about **\$301 higher** than Synthetic NJ (**RMSPE ≈ 65.55**).
- Forecasting projects NJ film spending from **\$874.7M in 2025 to \$1.12B by 2030** ($R^2 \approx 0.83$).
- Employment shifted from **+34.6 jobs per year before 2018 to -156.6 jobs per year after 2018** ($R^2 \approx 0.23$), showing a more volatile pattern than wages and establishments.

Acknowledgements:

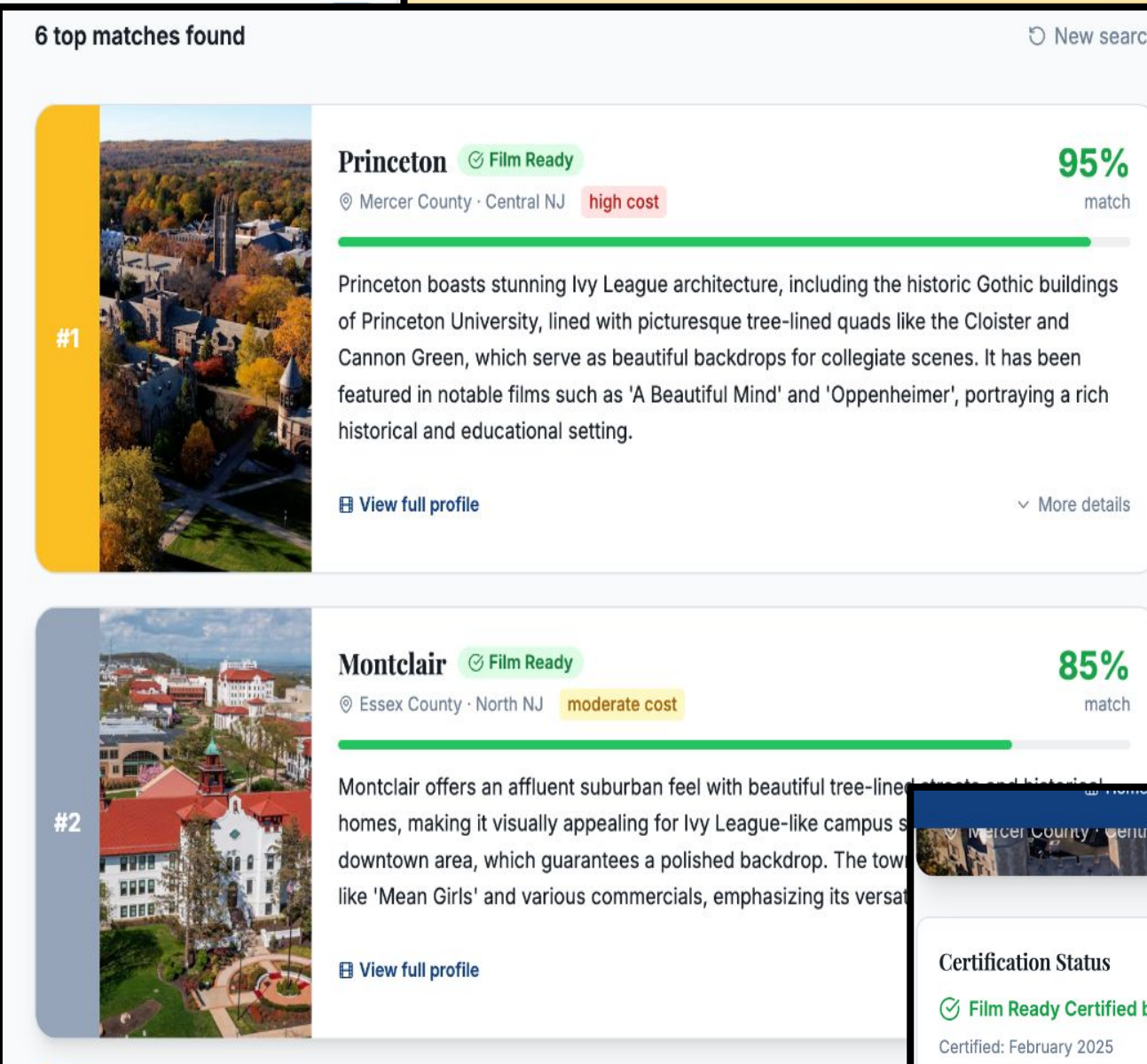
We would like to thank Dr. Antonios E. Marsellos and Dr. Emre Yetgin for their encouragement.

AI Location Scout: Turning Data Into Decision Support

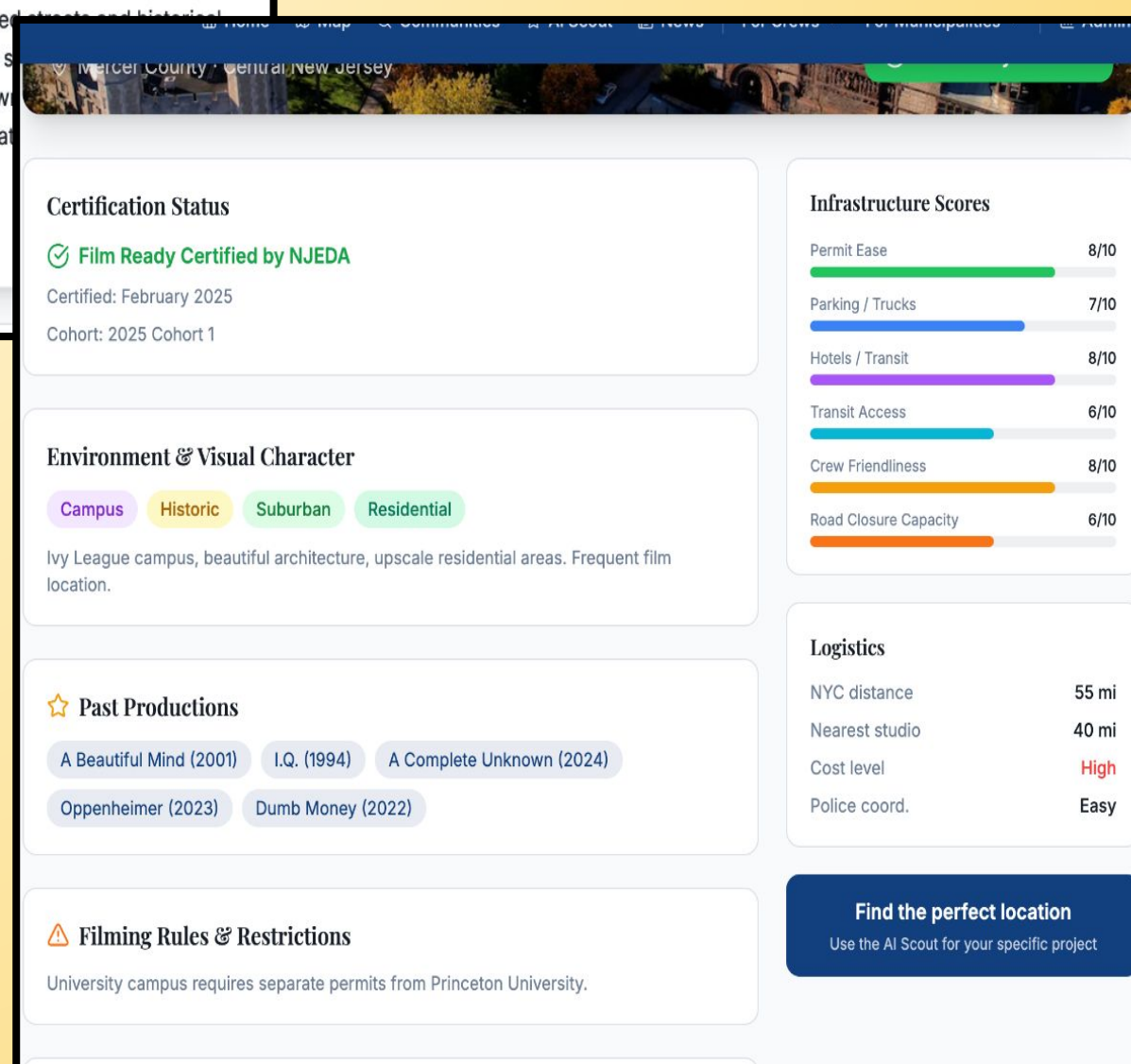


Producers enter filming needs such as setting type, logistics, accessibility, and community readiness.

Prompt: "Ivy-League campus with grand stone buildings and tree-lined quads"



Location Scout recommends Princeton, a film ready communities that best matches the productions needs.



Location Scout explains why the community fits, helping producers compare options more quickly and fairly.

- Helps producers identify NJ film ready locations
- Allows municipalities to market themselves
- Ethically elevates smaller municipalities instead of exclusively recommending the most commonly used locations.

Figure 4: AI Location Scout interface showing search inputs, ranked community matches, and decision-support details for production planning.

AI Location Scout Development

- Built as an AI recommendation system that combines a pre-trained LLM with rule-based filters and similarity scoring.
- Google Maps API** token used for live geocoding, location searches, and measuring distances to key infrastructure.
- Trained on New Jersey film industry terms, local infrastructure stats, and official Film Ready certification data.
- Uses **similarity scoring** to Generate a match percentage by comparing a producer's exact needs (like budget, crew size, and setting) directly against a community's actual attributes.
- Programmed to prevent **algorithmic bias** by ignoring how "popular" a location historically is, ensuring smaller, overlooked towns get a fair chance to be recommended.

Discussion

- The rapid addition of new establishments proves the state's film industry is becoming a permanent ecosystem.
- A First Assistant Director provided real world context for this project, stating the tax incentive is "99% the reason" productions now choose New Jersey (student's interview). He also confirmed that while direct crew jobs fluctuate, local restaurants and hotels see economic spillovers.
- This exact gap motivated the creation of Location Scout. To ensure the projected \$1.12 billion in film spending translates into steady jobs, our ethical AI actively matches productions with smaller, newly certified communities. This democratizes visibility and influences statewide economic development.
- Job creation was temporarily delayed by global shocks like COVID-19 and recent labor strikes, the state's financial data proves the industry remained resilient despite these short-term disruptions.

Conclusion

- Models confirm structural, long-term growth in New Jersey's film wages and establishments, projecting state production spending will reach \$1.12 billion by 2030.
- Applied AI Solution: Location Scout translates these findings into a practical decision-support platform for producers, municipalities, developers, and policymakers.
- Equitable Ecosystem: By prioritizing ethical AI design, the tool prevents algorithmic bias and democratizes visibility for overlooked towns, fostering an accessible and sustainable ecosystem for statewide job creation.

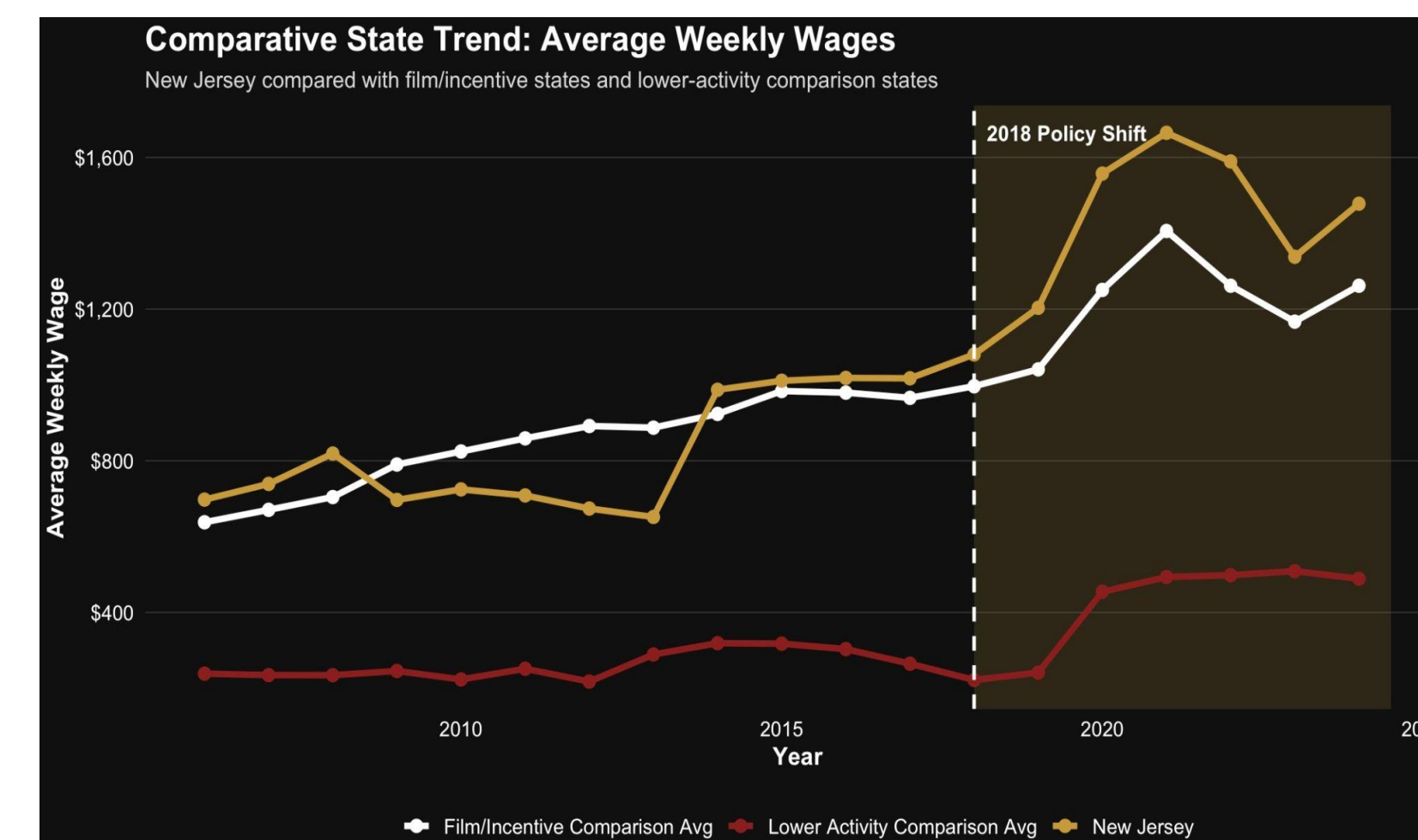


Figure 1: Comparative State Trend: Average Weekly Wages

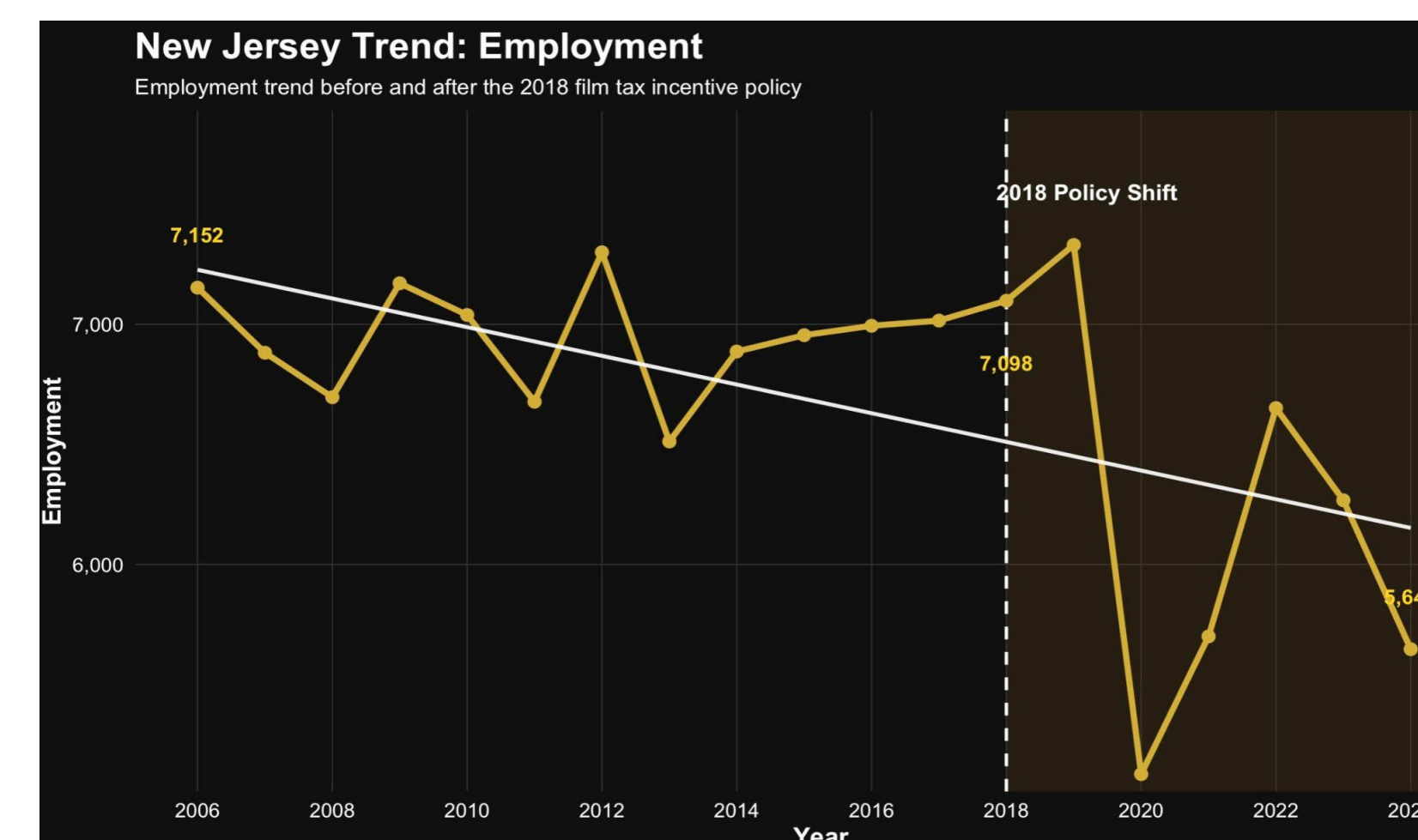


Figure 2: New Jersey Employment Trend

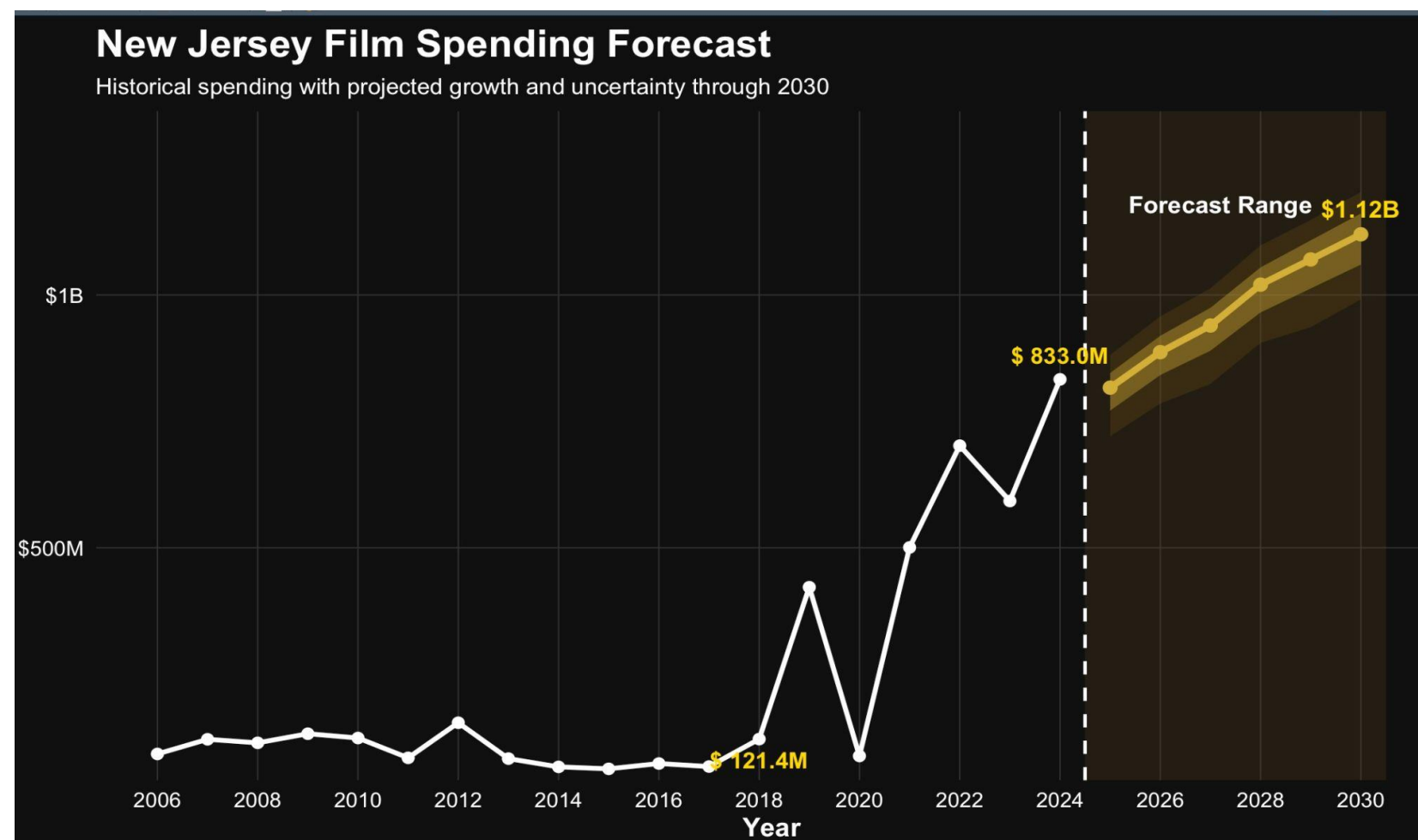


Figure 3: New Jersey Film Spending Forecast

Credit Authorship Contribution Statement

Bacchetta, D.; Lockhart, C.; **McCall Jr, P.:** Implementation, Writing, RStudio coding, Poster design.
Marsellos, A.E.: Supervision, Project administration, Guidance, Editing.
Yetgin, E.: Supervision, Project administration, Guidance, Editing.

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